FROST & SULLIVAN

Market Engineering

Market Study of China's Spatial Design Software Industry

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For and on behalf of
Frost & Sullivan (Beijing) Inc., Shanghai Branch Co.

Name: Terry Tse
Title: Consulting Director

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Overall Assumptions and Methodology

Methodology:

In preparing the report, Frost & Sullivan has relied on the statistics and information obtained through primary and secondary research.

•Primary research includes interviewing industry insiders, competitors, downstream customers and recognized third-party industry associations.

•Secondary research includes reviewing corporate annual reports, databases of relevant official authorities, as well as the exclusive database established by Frost & Sullivan over the past decades.

Assumptions:

The market size and forecasts were modeled by Frost & Sullivan based on the following assumptions.

•The social, economic and political conditions in China and main markets globally discussed will remain stable during the forecast period;

•Government policies on spatial design software industry and related downstream markets will remain consistent during the forecast period;

•The spatial design software industry will be driven by the factors which are stated in this report.

Research Scope

Market Study of Spatial Design Software Industry

Research Period

- Base year: 2024
- Historical: 2020 to 2024
- Forecasting: 2025E to 2029E

Service Market Scope

- Spatial Design Software Industry
- CAX Industry

Geographical Scope

- China
- Global

Content

Chapter	Section
1	Analysis of Key Downstream of Spatial Design Software Industry
2	Analysis of Spatial Design Software Industry
3	Competitive Landscape of Spatial Design Software Industry in China

1. Analysis of Spatial Design Software Industry

Analysis of Spatial Design Software Industry

Definition and Classification of General-purpose Design and Visualization Software Industry

- Design and visualization refer to the process of creating and representing ideas, concepts, or structures in a visually understandable and engaging way. This often involves using tools and techniques to transform abstract ideas into clear, visual representations, such as sketches, 3D models, or interactive environments. These tools are widely applied across industries like advertising, entertainment (including gaming and media), architecture, and product development, enabling businesses to create immersive, interactive visuals that enhance customer experiences, streamline product design, and improve marketing strategies. The global general-purpose design and visualization software market encompasses a wide range of applications, including spatial design, CAE, CAM, CAA, game design, advertising design and other visualization tools (including AR, VR and others). This market is driven by technological advancements in AI, cloud computing, and real-time rendering, as well as increasing demand for digital transformation across industries such as manufacturing, construction, healthcare, and entertainment. The market is characterized by high concentration in spatial design and fragmentation in other emerging verticals. While global players leverage historical dominance, Chinese vendors are gaining traction through policy support and niche solutions.
- The landscape of general purpose design software encompasses a range of specialized tools, each serving distinct yet sometimes overlapping functions. Spatial design software focuses on the aesthetic and functional aspects of space planning, primarily used in interior design and architectural visualization. It emphasizes user experience and interactivity, catering to designers and architects, and leveraging CAD capabilities. However, while CAD also serves for engineers and designers in industrial, mechanical, electronic, and broader architectural applications, spatial design remains focused exclusively on space-related solutions.

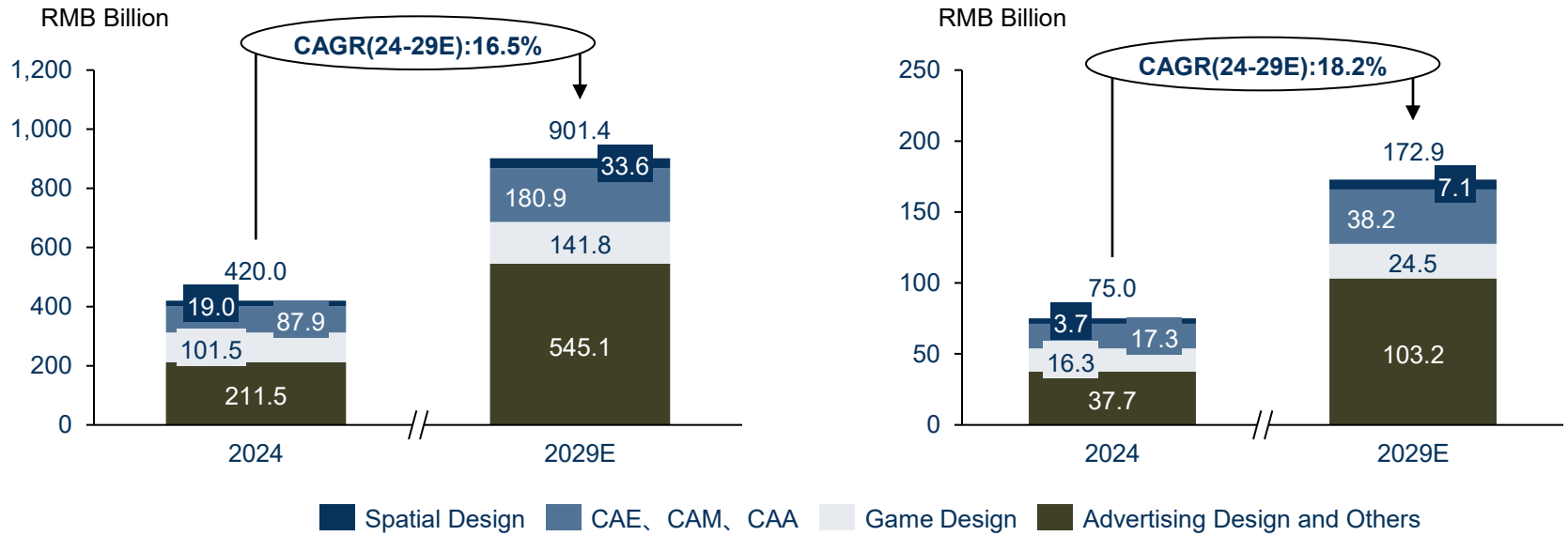
General-purpose Design and Visualization Software Market	Definition	Major Players
Spatial Design Software	• Tools for architectural, interior, and urban planning design, enabling 3D modeling, rendering, and virtual walkthroughs.	• The spatial design software market is highly concentrated, with major players including Kujiale, Sunvega, Yfang, Autodesk and etc.
CAE、CAM、CAA Software	• CAE refers to computer-aided engineering, CAM refers to computer-aided manufacturing, CAA refers to component application architecture.	• The global CAE, CAM CAA markets are moderately concentrated, with major players including Dassault Systèmes, Siemens, Autodesk, PTC and etc.
Game Design and Development Tools	• Software for 3D modeling, animation, and game engine development.	• The game design and development tool market is relatively decentralized, with major players including Unity, UNREAL, EPIC and etc.
Advertising Design Software	• Tools specialized for creating promotional content, brand materials, interactive advertisements, and marketing campaigns with advanced design, layout, and visualization capabilities.	• The advertising design software market is moderately diversified, with major players including Adobe, Figma, Canva, and Google Web Designer, etc.

Source: Frost & Sullivan

Analysis of Spatial Design Software Industry
Analysis of the General Purpose Design and Visualization Software Industry

- The global general purpose design and visualization software market is expected to reach RMB 901.4 billion by 2029, expanding from RMB 420.0 billion in 2024 at a CAGR of approximately 16.5%. Meanwhile, China's market is forecast to grow from RMB 75.0 billion in 2024 to RMB 172.9 billion in 2029, representing a higher CAGR of 18.2%. This strong growth in both global and Chinese markets reflects sustained demand for more advanced digital design solutions, driven by increasing adoption across diverse industries such as architecture, manufacturing, and media, as well as continued investment in digital transformation initiatives. These trends are accelerating the adoption of design and visualization software tools across sectors, reshaping how companies approach design, branding, and user engagement.
- Regarding market trends, the Group has shown strong competitiveness in the spatial design segment. Through continuous innovation and strategic enhancements, the Group's spatial design solutions have experienced steadily rising market share and penetration, reinforcing its leadership position in this niche. However, in the broader CAX market, the Group's current penetration remains relatively limited, as these segments are still largely controlled by the dominant international firms. Nevertheless, there is considerable potential for future growth in this area, especially as digital transformation accelerates and domestic customers increasingly seek localized, cost-effective alternatives. Finally, the general purpose design and visualisation software market operates in a separate competitive sphere, where the Group's products do not directly compete with those designed for sectors like gaming and animation design. The Group's focused strategy in spatial design, combined with targeted efforts to expand in the CAX market, will drive improved market share and sustained growth in the coming years.

Market Size of General Purpose Design and Visualization Software Industry, Global & China, 2020-2029E



Source: Frost & Sullivan, China Software Industry Association

Analysis of Spatial Design Software Industry
Definition and Classification of Spatial Design Software Industry

- The spatial design software refers to the software tools and related digital platforms used for creating and managing residential, commercial and other spaces, of which CAD software plays a crucial role. Spatial design software allows users to create complex geometric shapes and dynamic effects, visualize their design works and performs modeling tasks.

Source: Frost & Sullivan

Analysis of Spatial Design Software Industry
Development History of China's and Global Spatial Design Software Industry

- The evolution of the spatial design software industry, both globally and in China, reflects a progression from basic CAD tools in the early stages to the current trends of being cloud native, intelligent and AI empowered. Globally, technological advancements and business scenarios innovation have driven continuous improvements in software functionality and performance.
- The further integration of artificial intelligence (AI) and big data technologies, on the one hand, will continue to propel the spatial design software industry towards greater intelligence and productivity. On the other hand, visualization technology is also evolving, enabling experimentation and enhancing comprehension in residential, commercial and industrial design applications by providing intuitive and immersive representations of designs and simulations.

Development History of Global and China's Spatial Design Software Industry

Phase 1: Early CAD Tools Stage

①
In the early stages, spatial design software was primarily based on **Computer-Aided Design (CAD)** technology, focusing on 2D drafting and simple 3D modeling.

Phase 2: Emergence of the BIM Era

②
From 2000 onwards, **Building Information Modeling (BIM)** technology began to emerge, driving significant transformations in the global spatial design software industry.

③
By around 2010, the world had entered the BIM era, with many software vendors introducing user-friendly BIM tools that became foundational elements in the architecture and construction industry.

Phase 3: AI and Virtual Space Era

④
Since the 2010s, **AI technology and virtual reality (VR) and augmented reality (AR)** technologies have gradually integrated into spatial design software.

⑤
As visualization technology evolves, spatial design software is exploring new applications in industrial design, with key players taking the lead.

Global

1960s-1970s | **2000s** | **2010s** | **Future**

China

Phase 1: Initial Stage

①
During the early stages of the reform and opening-up period, China mainly relied on foreign software such as Autodesk Revit.

Phase 2 : Localization Stage

②
From 2000 to 2010, China began focusing on localized development. The China Academy of Building Research launched PKPM Building Information Modeling System (PBIMS), the first domestically developed BIM system. Some companies made notable progress in the 3D modeling field by connecting visualization technology.

Phase 3: Rapid Growth Stage

③
China's spatial design software market has seen remarkable growth, with **cloud-based solutions significantly outpacing traditional on-premise software**.

④
Local companies are holding more market share and maintaining a fast growth rate. They are actively scoping out AI solutions incorporating visualization techniques to enhance their competitive edge.

Source: Frost & Sullivan

Analysis of Spatial Design Software Industry
Analysis of Spatial Design Software Empowerment

- Spatial design software plays a pivotal role in empowering the design process by enhancing key aspects. Functionality is improved as automatic design tools harness robust algorithms and databases, expediting scheme creation and providing users with a wealth of design options. Cross-business collaboration is also strengthened through AI-powered features that seamlessly translate design concepts into production-ready drawings, while real-time, multi-terminal interaction fosters seamless teamwork. Quality control is reinforced by high-precision rendering, allowing real-time realistic previews, material adjustments, and ensuring designs meet standards, boosting satisfaction and reliability.

Spatial Design Software Empowerment

Function Applicability Enhancement

- **Design Efficiency:** Design automation leverages robust algorithms and extensive databases to enable the automatic design schemes, helping users quickly complete design plans and providing them ample design inspiration.
- **Modeling & Visualization Efficiency:** The automatic design feature fosters swift creation and visualization of spatial design models, especially by leveraging the GPU hardware and advanced rendering technology, offering users intuitive design comprehension and minimizing revision time.

Collaboration Enhancement

- **Integrated Monitoring:** Fully-streamlined applications of spatial design software enable enterprises to holistically manage design content and oversee the entire order–design–manufacture process, facilitating post-design collaborations with manufacturers and other related parties.
- **Real-Time Interaction:** The multi-terminal viewing feature supports real-time interaction, such as simultaneous review and editing, fostering enhanced productivity through collaboration, further streamlining the design process.

Quality Control Enhancement

- **Rendering Precision:** The advanced rendering engine of the spatial design software enables users to produce stunningly realistic visuals instantly. Real-time material previews and seamless adjustments guarantee that the final outcome surpasses expectations, elevating user satisfaction.

Source: Frost & Sullivan

Analysis of Spatial Design Software Industry
Analysis of the Empowerment and Development Impact of AI

- AI technologies have been significantly empowering the spatial design software industry, leading to unprecedented levels of innovation, efficiency, and optimization. GPU and AI rendering advancements optimize rendering processes and enhance user experiences, ensuring smooth and high-performance graphics operations, thereby driving industry innovation.

GPU Hardware

- **Empowerment**
  - **Optimized Rendering:** The efficient parallel computing capabilities of GPU clusters enable spatial design software to allocate computing resources optimally for graphics rendering tasks of varying scales and complexities, achieving high-performance rendering.
  - **User Experience:** High-performance GPU clusters support high-resolution displays and complex graphic operations, which ensure a smooth operating experience with less latency when using spatial design software.
- **Development**
  - **Industry Innovation:** High-performance GPU clusters provide the computational support that enables spatial design software to increase its capability and drive innovation by integrating AI functions.

AIGC Software

- **Empowerment**
  - **Design Efficiency:** AIGC technology can automatically generate multiple design options based on simple and descriptive instructions, for users to choose based on sketches or photos, which greatly improves users' design efficiency.
  - **Creativity Inspiration:** AIGC can analyze a large amount of design data and suggest design ideas tailored to each user's unique taste, which empowers users to find inspiration for their projects by exploring and discovering a wide range of design themes and new elements.
- **Development**
  - **Design Transformation:** AI copilot streamlines the design process by automatically converting uploaded files into immersive 3D schemes. This advanced toolset offers a seamless integration of personalized styles, fostering rapid iteration and a unique creative edge.

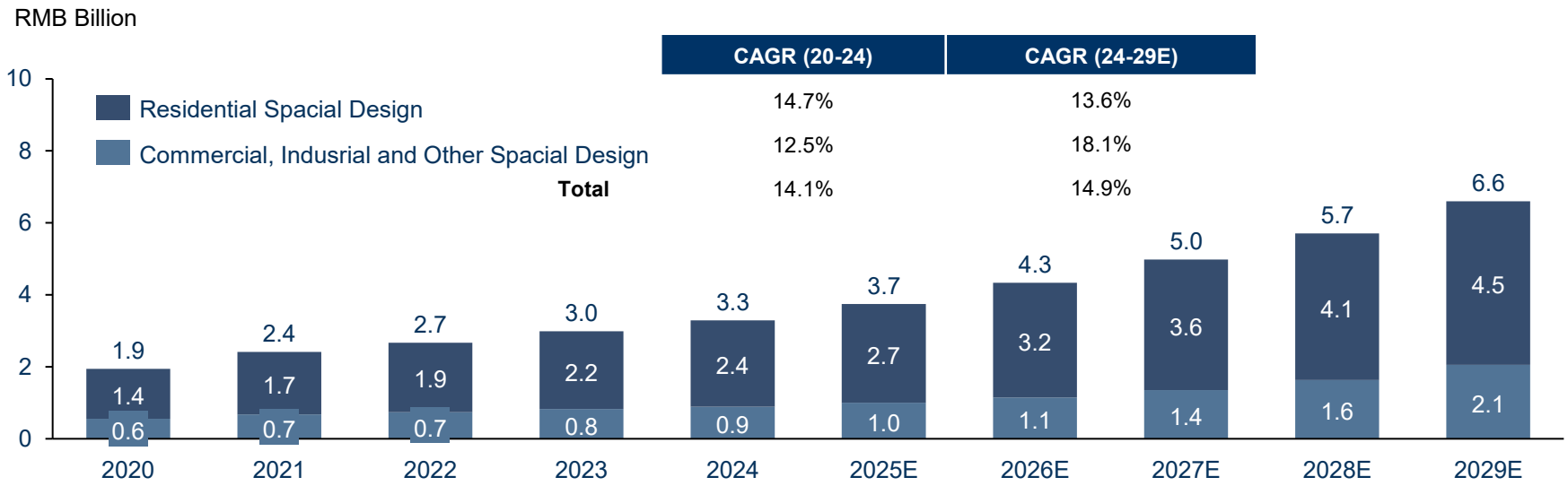
Source: Frost & Sullivan

Analysis of the Spatial Design Software Industry
Market Size of China's Spatial Design Industry (1/2)

- Spatial design software is a type of digital tool used for creating, editing, and analyzing spatial layouts, widely applied in fields such as architectural design, interior design, urban planning, and infrastructure architecture. The user demand for residential, commercial, and industrial spaces has continued to evolve, extending beyond functionality and usability to esthetics, quality and convenience. The increasingly sophisticated demand has been heightening the standard for spatial design. Designers and architects now need more powerful and intelligent tools for innovative, high-quality solutions, boosting demand for spatial design software. Additionally, government support for smart buildings and green architecture is further fueling the growth of the spatial design software industry by setting more requirements for spatial design works.

- China's spatial design software industry has seen substantial growth over the last five years, rising from RMB 1.9 billion in 2020 to RMB 3.3 billion in 2024, representing a CAGR of 14.1%. It is projected to further increase to RMB 6.6 billion by 2029, representing a CAGR of 14.9% from 2024 to 2029.

Market Size of China's Spatial Design Software Industry, by Revenue, 2020-2029E



	CAGR (20-24)	CAGR (24-29E)
Residential Spacial Design	14.7%	13.6%
Commercial, Indusrial and Other Spacial Design	12.5%	18.1%
Total	14.1%	14.9%

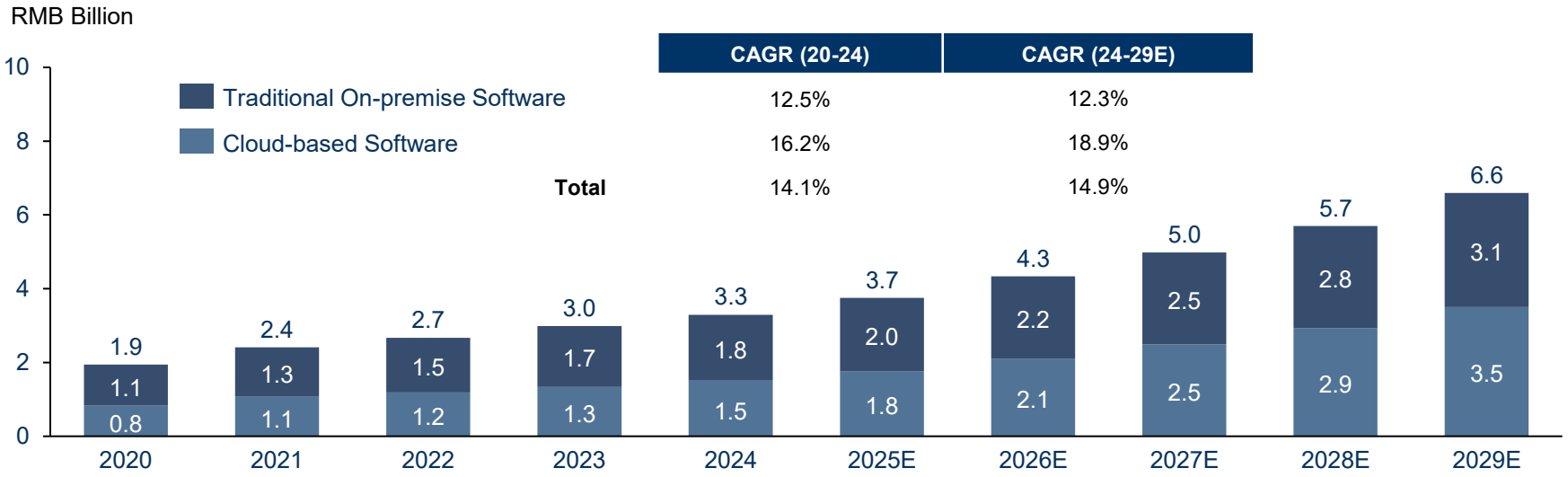
Source: Frost & Sullivan, National Bureau of Statistics of China

Analysis of the Spatial Design Software Industry
Market Size of China's Spatial Design Industry (2/2)

- China's spatial design software market can also be categorized by deployment mode into two categories: traditional on-premise software and cloud-based software. Traditional on-premise software refers to software deployed on local servers or devices, requiring users to install and operate it via local hardware infrastructure. Cloud-based software refers to software delivered through cloud computing platforms, enabling users to access and use functionalities via the internet without local installation.

- In 2020, the total market size was RMB1.9 billion, with traditional on-premise software contributing RMB1.1 billion and cloud-based software RMB0.8 billion. By 2024, the market expanded to RMB 3.3 billion — traditional on-premise software reached RMB1.8 billion, and cloud-based software grew to RMB1.5 billion — supported by a 12.5% CAGR for the former and 16.2% for the latter during 2020-2024, highlighting stronger momentum in cloud solutions. The market is projected to reach RMB6.6 billion by 2029, with traditional on-premise software at RMB3.1 billion and cloud-based software at RMB3.5 billion, with a CAGR of 12.3% and 18.9%, respectively.

Market Size of China's Spatial Design Software Industry, by Revenue, 2020-2029E

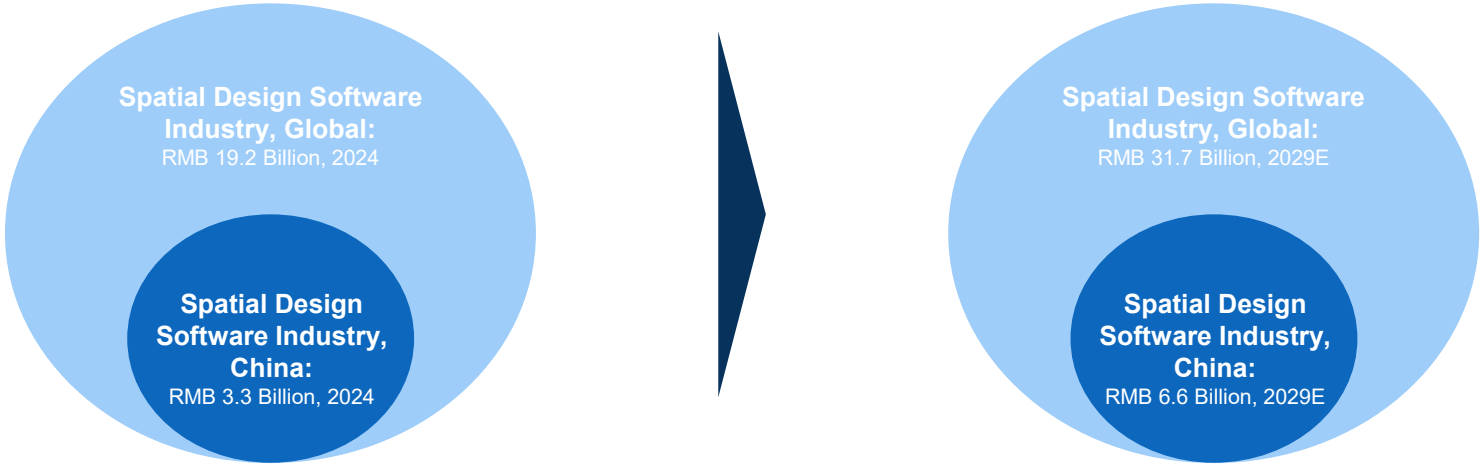


Source: Frost & Sullivan, National Bureau of Statistics of China

Analysis of the Spatial Design Software Industry
Growth Potential of Global and China's Spatial Design Industry

- The market size for spatial design software in China is projected to grow substantially, driven by increasing demand for innovative design solutions across various business scenarios. Similarly, the global market for spatial design software is poised for significant growth, with growing demand across diverse countries and regions.
- In recent years, China's spatial design software industry has undergone remarkable growth, fueled by the increasing demand for spatial design solutions across various sectors. By 2024, the market size had reached RMB 3.3 billion, and it is projected to expand significantly, reaching RMB 6.6 billion by 2029. On a global scale, the spatial design software industry represented a market size of RMB 19.2 billion in 2024. Furthermore, as the digital transformation continues to sweep across the globe, with the growing emphasis on productivity and efficiency in design, the global market will continue to expand, reaching RMB 31.7 billion by 2029.
- In 2024, the global spatial design software market showcased notable segmental distinctions. For industry verticals, the share of commercial, industrial and other design segment in the global market was approximately 35%—a higher proportion compared to China. In terms of deployment models, cloud-based software accounted for around 55% of the global market, surpassing China's share of cloud-based solutions. These figures highlight that globally, cloud-based software and the commercial/industrial/other design verticals held a more dominant position in 2024 compared to the Chinese market.

Growth Potential of China's Spatial Design Software Industry, 2024 & 2029E



Source: Frost & Sullivan, National Bureau of Statistics of China, International Monetary Fund

Analysis of the Spatial Design Software Industry
Analysis of Key Drivers of China's Spatial Design Software Industry

Driving Factor of China's Spatial Design Software Industry

Growing Demand from Various Scenarios

- The market for spatial design software is primarily driven by the increasing demand from downstream industries such as architecture, urban planning, and real estate development. As these sectors continue to grow, there is a rising need for sophisticated tools that can accurately model, visualize, and optimize spatial layouts and designs. The shift towards sustainable and efficient design practices, coupled with advancements in technology like AI and VR , has further heightened the demand for spatial design software. These tools enable designers to create more precise, cost-effective, and innovative solutions, meeting the evolving needs of clients and stakeholders in these industries. Moreover, the manufacturing sector, particularly in automotive, aerospace, and electronics, drives the need for precise design and manufacturing processes, which could be the potential future development direction for spatial design software companies.

Industry Globalization

- As companies expand their operations across borders, there is an increasing need for standardized tools that can accommodate diverse geographical, cultural, and regulatory requirements. Spatial design software that offers multi-language support, seamless integration with international building codes, and compatibility with various regional design standards is becoming essential. Moreover, the rise of global collaborations and remote work has amplified the demand for cloud-based solutions that allow design teams from different regions to collaborate in real-time. For example, in a scenario where a design team from China and a design team from South Korea collaborate for a commercial design project, the cloud-based spatial design software allows them to collaborate in their native language, driving the adoption and growth of spatial design software on a global scale.

Product Customiza-tion and Flexibility

- Customization and flexibility are important market drivers for spatial design software, as they help meet the evolving needs of users across industries. In a setting where project requirements differ widely, software that offers customized templates, adjustable parameters, and adaptable workflows gains added value. Design tools enable users to modify and scale their projects with ease, catering to specific client demands. This adaptability not only enhances user satisfaction but also positions spatial design software as a preferred solution for businesses looking to maintain a competitive edge through innovative and customized solutions.

Thriving Virtual Space for Various Business and Entertainment Scenarios

- As industries undergo digital transformation, businesses are increasingly tapping into virtual environments to facilitate immersive marketing and customer engagement. Sectors within the entertainment industry, including gaming, virtual reality, and live-streaming, are fostering a thriving ecosystem that underscores the necessity for sophisticated design tools.

Source: Frost & Sullivan

Analysis of the Spatial Design Software Industry
Development Trends of China's Spatial Design Software Industry

1 Increasing Prevalence of Cloud-Based Solutions	• The shift towards cloud-based platforms is transforming the spatial design software landscape. Compared to on-premise installed software, cloud-based software has minimized requirements for local hardware configurations and does not trap users at their local workstations, enabling greater efficiency and flexibility. Cloud-based solutions offer greater accessibility, allowing teams to collaborate in real-time from different locations. Additionally, cloud solutions provide scalable resources, making it easier for companies to manage complex projects without the need for extensive upfront on-premise infrastructure.
2 Growing Demand for Interoperability	• As design projects become more complex and multidisciplinary, there is an increasing need for spatial design software that can seamlessly integrate with other tools and platforms such as CAM software. Interoperability allows different software systems to communicate and share data effectively, facilitating smoother collaboration among architects, engineers, and contractors. This trend is driving the development of software solutions that support a wide range of file formats, application programming interfaces (APIs), and plug-ins, enabling users to work across multiple platforms without losing data integrity or workflow efficiency.
3 Improvement of High-quality 3D Rendering Technology	• With improved computing capabilities and advances in graphics technology, spatial design software in China has made significant improvements in 3D rendering. Advanced rendering engines and real-time rendering techniques such as physics-based rendering and ray tracing allow designers to create more realistic and detailed visual effects. These technologies not only improve the realism of the design scheme, but also accelerate the rendering speed and improve the work efficiency.
4 Widespread Adoption of Virtual Reality (VR) and Augmented Reality (AR)	• The rapid development of VR and AR technologies has enabled designers to create immersive spatial experiences. Through these technologies, the client can "step into" the design proposal in real time in the virtual environment to view the spatial layout and decorative outputs. This highly interactive visualization greatly enhances the intuitiveness of the design display and the sense of customer participation, and reduces the cost of modification and decision-making.
5 Technology Advancements	• Technology advancements are trending in the spatial design sector. Traditional industry procedures are failing to match the design needs and preferences of millennials and tech-savvy generations who are increasingly making decisions for their homes, workplaces, and business environments. Designers must now produce solutions more quickly that meet both functional and aesthetic requirements. Because they are actively involved in the design process, this generation of end users requires designs that are simple to share and visualize. Only digitalization using cutting-edge software technology can meet this need. A revolutionary change in spatial design software has been made possible by technological developments like the development of AI and hardware configuration iterations.

Source: Frost & Sullivan

2. Competitive Landscape of Spatial Design Software Industry in China

Competitive Landscape of Spatial Design Software Industry in China
Competitive Landscape Analysis

- The competitive landscape of China's spatial design industry is relatively concentrated, with over 50 players actively participating, yet a few key players dominate the market. Despite the presence of certain international competitors, which still hold a significant market share in China, the industry is witnessing a leaning towards domestic alternatives, particularly in the spatial design market. In China's spatial design software industry as a whole, the Company ranked first in 2024 in terms of revenue, representing a market share of 23.2%.

- The spatial design software market is characterized by high concentration both in the PRC and globally, with dominant players capturing significant market share. The similarities and differences between PRC and overseas markets lies in the technological adoption patterns. In PRC market, cloud-based spatial design software dominates, driven by rapid digital transformation across industries, favorable policies supporting SaaS adoption, and demand for collaborative, real-time design tools. In overseas markets, traditional on-premise solutions remain prevalent, particularly in mature markets such as North America and Europe.

Source: Frost & Sullivan

Competitive Landscape of Spatial Design Software Industry in China
Ranking of China's Spatial Design Software Industry (1/2)

Ranking of Spatial Design Software Industry, China, 2024

Ranking	Company	Revenue (RMB Billion)	Market Share (%)	Vertical Coverage
1	**The Company**	**0.8**	**23.2%**	Residential, commercial, industrial and others
2	Company A [1]	0.7	21.3%	Architecture, engineering, construction (AEC), manufacturing, and media/entertainment
3	Company B[2]	0.4	11.7%	Industrial manufacturing and infrastructure
4	Company C[3]	0.3	8.6%	Customized home furnishing and cabinetry design
5	Company D[4]	0.1	3.1%	Customized home furnishing and kitchen/bathroom design

Source: Frost & Sullivan

Competitive Landscape of Spatial Design Software Industry in China
Ranking of China's Spatial Design Software Industry (2/2)

Notes:

(1) Company A is a U.S. company founded in 1982. It is a global leader in 3D design, engineering and entertainment technology solutions, spanning architecture, engineering, construction, product design, manufacturing, media, and entertainment. Its comprehensive software suite serves diverse industries, including architecture, construction, manufacturing, and entertainment, with tools for Building Information Modeling (BIM), generative design, and cloud-based collaboration platforms.

(2) Company B is a listed Chinese company founded in 1998. It is a leading provider of All-in-One CAx (such as CAD/CAE/CAM) solutions. Its proprietary platform supports mechanical, electrical, and electronic design automation, catering to manufacturing, automotive, and aerospace sectors with localized compliance and costeffective licensing models.

(3) Company C is a Chinese company founded in 2013. It is a leading provider of digital solutions for the home furnishing industry. Its solutions combine AI-powered 3D modeling, VR visualization, and supply chain optimization tools, enabling customized interior design workflows for residential and commercial clients.

(4) Company D is a Chinese company founded in 1994. It is a leading provider of the home furnishing industry. Its platform streamlines design-to-production processes, emphasizing modular furniture systems and cross-device compatibility for hybrid living and workspaces.

(5) The ranking caliber of the Chinese companies involved in the table is the company's total revenue.

Source: Frost & Sullivan

Competitive Landscape of Spatial Design Software Industry in China
Entry Barrier Analysis

1 Technological Barriers

- The spatial design software industry has significant technological barriers due to the complexity and specialization required in developing advanced technologies. High-speed 3D rendering technologies require a complex hybrid parallel computing approach, combining the strengths of GPUs to optimize image quality and processing speed. The integration of AI applications, such as automated interior design systems and environment understanding engines, further elevates the technical complexity.

2 Capital Barriers

- The extensive investments required in hardware, software, and R&D represent a significant capital barrier for new entrants. Establishing high-performance infrastructure needs substantial spending on advanced GPUs to support high-speed rendering and parallel computing. Additionally, significant resources must be allocated to develop advanced algorithms and technologies, such as cloud-native 3D graphics and AI-driven applications. Moreover, ongoing research and development are crucial for optimizing and advancing these technologies.

3 Talent Barriers

- The spatial design software industry has notable talent barrier, given the specialized skills and expertise required to drive innovation and maintain technological leadership. The competition for such talent is intense, as professionals with these skills are highly sought after across multiple tech sectors, including software development, AI research, and computational graphics.

4 Brand Barriers

- Setting up a strong brand presence in the spatial design software industry represents a substantial entry barrier. Building and protecting a brand involves developing and leveraging intellectual property assets, including patents, trademarks, and copyrights. New entrants must navigate the competitive landscape dominated by established players with strong brand recognition and market presence, while the established reputation and customer loyalty of existing players present a formidable competitive barrier.

Source: Frost & Sullivan

Verifications

- The competitive landscape of China's spatial design software market is relatively concentrated, with the top three players accounting for over 50% of the market share. International competitors such as Autodesk, Inc. ("Autodesk") hold a significant share of the market; however, in recent years, a clear trend of domestic substitution has emerged. Chinese companies, including the Issuer and Guangzhou Zhongwang Longteng Software Co., Ltd. ("ZWSOFT"), together command a market share of approximately 33.0%. Among them, the Issuer has achieved a 23.2% market share, marking an initial realization of domestic substitution in China's spatial design sector.

- Autodesk was founded in 1982 and is headquartered in California, United States. Its principal business is vertical industry design software. The company is listed on the NASDAQ under the ticker symbol "ADSK". It primarily offers four major product lines: Architecture, Engineering and Construction (AEC); AutoCAD and AutoCAD LT for 2D & 3D design (general-purpose industrial design software); Product Design and Manufacturing (MFG); and Media and Entertainment (M&E).

- Autodesk's Competitive Advantages vs. the Issuer
  - **Product**: AutoCAD, launched by Autodesk in 1982, is widely applied across various domains. In contrast, the Issuer focuses on spatial design software, enabling a deeper understanding of and alignment with user needs.
  - **R&D Investment**: The Issuer continuously drives innovation through research and development, maintaining an industry-leading R&D expense ratio of approximately 45–73% over the past three years, compared to Autodesk's 24–25%.
  - **Regional Advantage**: With a strong presence in China, the Issuer (Kujiale) ranked No.1 in the Chinese market in 2024, offering products that better cater to local user preferences and requirements. Autodesk, by comparison, does not prioritize the Chinese market, with the Asia-Pacific region accounting for only 18% of its revenue.
  - **User Friendliness**: The Issuer provides a simple, intuitive interface with a low learning curve, enabling non-professional users to adopt its software quickly. Autodesk, however, primarily targets professional designers in industrial and architectural fields, where adoption often requires longer training and more complex operations.

Source: Frost & Sullivan

Thank You

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